

SOLÈNE PETTIER

Senior Smart Contract / Protocol Engineer

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SUMMARY

Senior Smart Contract and Protocol Engineer with 5+ years designing, building, and securing production blockchain systems — including protocols managing \$18M TVL, \$1.5M in monthly reward distributions, and cross-chain infrastructure spanning 10 Kadena chains. Deep expertise in Solidity, Foundry, EVM architecture, DeFi protocols, and upgradeable smart contracts, with a strong security background spanning threat modeling, audit preparation, fuzz/invariant testing, and incident response to production exploits.

CORE SKILLS

- **Smart Contracts:** Solidity, EVM, Foundry, Hardhat, OpenZeppelin, Diamond Proxy, Upgradeable Contracts, Gas Optimization, Assembly
- **Decentralized Protocols:** DeFi, DEXs, Perpetuals, Yield Strategies, Staking, Tokenomics, Oracles, Insurance Protocols, DePIN
- **Security:** Threat Modeling, Smart Contract Security, Audit Preparation, Vulnerability Remediation, Fuzz Testing, Invariant Testing, Exploit Response, Bug Bounties
- **Blockchain Infrastructure:** Cross-Chain Systems, Hyperlane, Pact, Safe Multisig, Multi-Chain Deployments
- **Software Engineering:** TypeScript, JavaScript, Node.js, Python, Ethers.js, CI/CD, Docker, Kubernetes

PROFESSIONAL EXPERIENCE

IMPOSSIBLE CLOUD NETWORK — Senior Blockchain Engineer

Mar 2025 – Present | Remote

DePIN protocol for decentralized cloud storage.

- Designed and implemented production Solidity smart contracts for provider registration, payments, and reward distribution using a modular, upgradeable Diamond architecture.
- Implemented on-chain tokenomics supporting approximately \$1.5M in monthly reward distributions.
- Owned the smart contract security lifecycle, including threat modeling, audit preparation, vulnerability remediation, and resolution of 15+ post-launch bug bounty reports.
- Built migration scripts and deployment tooling for contract upgrades and protocol state transitions.
- Automated governance and maintenance workflows using Safe Multisig and the Safe SDK.
- Built CI/CD pipelines for automated smart contract testing and deployment.
- Deployed OpenZeppelin Monitor on Kubernetes to watch on-chain events and alert on failures via Slack/email.
- Authored NatSpec documentation, architecture diagrams, and technical specifications for engineering and audit teams.

TECHNOLOGIES: Solidity, Foundry, OpenZeppelin, EVM, TypeScript, Node.js, Safe, Kubernetes, CI/CD, DePIN, DeFi

LINKS: ICN live dapp (console.icn.global)

IDEASOFT x KADENA — Blockchain Engineer

Jun 2024 – Jul 2025 | Remote | Contract

Kinesis Bridge: cross-chain bridge connecting Base and Kadena networks using Hyperlane.

- Built and deployed a mainnet bridge connecting Base to 10 Kadena chains through Hyperlane infrastructure.
- Implemented the Kadena-side Hyperlane messaging interface in Pact and developed TypeScript deployment automation for both ecosystems.
- Integrated Hyperlane's Interchain Security Module with validator-based message verification.
- Designed and maintained automated deployment infrastructure for approximately 30 token contracts across multiple networks.
- Developed security tests covering replay attacks, validator compromise, and message injection.
- Responded to a production exploit involving a malicious token integration by pausing the protocol and shipping a fix within 12 hours.
- Maintained NatSpec documentation, architecture diagrams, and technical specifications for auditors and engineering teams.

TECHNOLOGIES: Solidity, Pact, TypeScript, Node.js, Hyperlane, EVM, Docker, CI/CD, OpenZeppelin

LINKS: Kinesis Bridge source code ([GitHub](#))

IDEASOFT x JIBX — Blockchain Engineer

Apr 2024 – Jul 2024 | Remote | Contract

Perpetual DEX with a fully on-chain central limit order book.

- Built a fully on-chain order book using a red-black tree data structure and optimized storage operations.
- Implemented on-chain order matching, position management, and liquidations entirely in Solidity.
- Optimized gas usage and bytecode size through storage packing, inline assembly, and efficient data structures.
- Developed comprehensive unit, fuzz, and invariant test suites before mainnet deployment.
- Built TypeScript and JavaScript automation tooling for contract interaction and on-chain operations.
- Authored and maintained NatSpec documentation, architecture diagrams, and technical specifications for auditors and engineering teams.

TECHNOLOGIES: Solidity, Foundry, DeFi, Oracle Systems, Node.js, Ethers.js, Red-Black Trees

LINKS: Perpetual DEX source code ([GitHub](#))

IDEASOFT x DOLLET WALLET — Blockchain Engineer

Dec 2023 – Apr 2024 | Remote | Contract

On-chain yield strategies for a DeFi wallet.

- Architected a modular framework for vaults, yield strategies, oracles, governance, and fee management.
- Designed a reusable strategy integration framework with common lifecycle management and accounting.

- Implemented integrations with Ether.fi, Pendle, and Wombat.
- Built a swap-routing service supporting token swaps across 5 DEXs: Uniswap, PancakeSwap, Curve, Balancer, and SushiSwap.
- Implemented price validation using Chainlink, RedStone, and Uniswap TWAP with slippage protection.
- Automated multi-chain deployments using TypeScript-based deployment tooling.
- Authored and maintained comprehensive NatSpec documentation to support audits, code reviews, and internal development.

TECHNOLOGIES: Solidity, Foundry, TypeScript, EVM, Oracles, Node.js, Ethers.js, DeFi, DEX Routing

LINKS: Dollet Wallet on X

IDEASOFT x DEIN — Blockchain Engineer

Sep 2021 — May 2023 | Remote | Contract

Decentralized insurance protocol managing coverage, premiums, claims, staking, and yield strategies on-chain.

- Refactored and extended core insurance protocol contracts covering policy issuance, coverage, claims processing, and premium management.
- Led the on-chain migration of 20 insurance policies securing approximately \$18M TVL during the Bridge Mutual → DEIN rebrand.
- Designed customizable policy lifecycles covering submission, approval, active coverage, and closure.
- Built staking mechanisms for underwriting insurance coverage and governance voting.
- Implemented encrypted voting, reputation scoring, reward distribution, and leverage logic for risk allocation.
- Integrated yield strategies across Aave, Compound, and Yearn with invariant testing to verify principal preservation and yield behavior.
- Produced technical specifications and architecture documentation for cross-functional teams, and supported frontend integration of smart contracts with the dApp.

TECHNOLOGIES: Solidity, Foundry, Hardhat, OpenZeppelin, Ethers.js, DeFi, Insurance Protocols, Invariant Testing, Fuzz Testing, Staking, Yield Strategies

LINKS: DEIN live dapp (app.dein.fi) · DEIN on X

PERSONAL TECHNICAL PROJECTS

Atomic Decentralized Features

Collection of standalone Solidity contracts and scripts demonstrating core blockchain and DeFi primitives, including reward accumulators, AMM arbitrage, Merkle airdrops, reentrancy attack demonstrations, and multichain bridge mechanics.

TECHNOLOGIES: Solidity, Foundry, Hardhat, JavaScript, Python

LINKS: reward accumulator (GitHub) · amm arbitrage (GitHub) · reentrancy hack (GitHub) · merkle airdrop (GitHub) · multichain bridge (GitHub)

Escape Cockpit

Interactive 3D browser escape room built with React and Babylon.js. Players take on the role of a pilot and solve a sequence of observation, logic, and interaction puzzles — reading airport codes, computing an ETA, adjusting a runway heading, and entering a final landing code — before a four-minute timer runs out.

TECHNOLOGIES: React, TypeScript, Vite, Babylon.js, CSS, Claude

LINKS: Escape Cockpit source code (GitHub) · Escape Cockpit live demo (Netlify)

EDUCATION

ALYRA SCHOOL — Full Stack Blockchain Developer

Jan 2021 — Jul 2021 | Paris, France

Certification covering the full blockchain stack: Solidity smart contracts, Web3 integration, React front-ends, and Node.js back-end services. Included building and deploying dApps end-to-end.

LINKS: Alyra certificate verification (bcdiploma.com)

NATIONAL SUPERIOR SCHOOL OF MINES — MSc Information Technology Engineer

Sep 2013 — Jun 2016 | Albi, France

Three-year engineering degree from one of France's top engineering school, covering IT systems, architecture, networks, and software development.

LINKS: MSc diploma verification (cvtrust.com)

MICHELET HIGH SCHOOL — Intensive Mathematics & Physics Program

Sep 2011 — Jun 2013 | Vanves, France

Selective two-year program in advanced mathematics and physics: the standard French pathway before engineering schools. High-pressure, high-competition. Built the capacity for deep concentration and rigorous thinking that carries into every technical challenge since.